

1. **Introduction:** Object oriented design, history and advantages of object-oriented design (*Foundational*)
2. **Object Oriented Programming:** Terminology and features (*Foundational*)
3. Classes, objects (*Foundational — building blocks*)
4. Data encapsulation (*Encapsulation*)
5. Attributes and methods of a class, identification of attributes/data members of a class (*Encapsulation*)
6. Constructors (*Encapsulation*)
7. Types of constructors (*Encapsulation*)
8. Destructors (*Encapsulation*)
9. Access modifiers (*Encapsulation*)
10. Passing objects to function by value and by reference (*Encapsulation*)
11. this keyword in C++ (*Encapsulation*)
12. const vs non-const functions (*Encapsulation*)
13. Static data members & functions (*Encapsulation*)
14. Function overloading (*Polymorphism — compile-time*)
15. Function Overriding (*Polymorphism — run-time*)
16. Operator overloading (*Polymorphism — compile-time*)
17. Identification of classes and their relationships (*Foundational*)
18. Composition (*Inheritance/Class Relationships — "has-a"*)
19. Inheritance (*Inheritance — "is-a"*)
20. Multiple inheritance (*Inheritance*)
21. Protected access specifier in terms of inheritance (*Inheritance*)
22. Order of Constructor calling in child class and parent class (*Inheritance*)
23. Polymorphism (*Polymorphism*)
24. Abstract classes and interfaces (*Abstraction*)
25. Standard Template Library (Vector etc.) (*Additional C++ tool — not tied to a pillar*)
26. Exception Handling (*Additional C++ tool — not tied to a pillar*) (*check class slides and lab tasks for updated content*)